INCH-POUND
QQ-N-286H
July 6, 2026
SUPERSEDING
QQ-N-286G
December 7, 2000

FEDERAL SPECIFICATION

NICKEL-COPPER-ALUMINUM ALLOY, WROUGHT (UNS N05500)

The General Services Administration has authorized the use of this federal specification by all federal agencies, some states and localities.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers all nickel-copper-aluminum alloy wrought forms formerly classified as class A. This alloy is designated under the Unified Numbering System for Metals and Alloys as UNS N05500. Except for cases where nonmagnetic properties are paramount, this alloy is intended to be age hardened prior to placing the material in service.

1.2 Forms and conditions. The material should be furnished in the following forms and conditions, as specified (see 6.2).

a. Form 1 – Bar, Rod, Unaged (for bar and rod formed by forging, see 1.2.1 and form 7).
   (1) Condition HU: Hot finished, unaged.
   (2) Condition CU: Cold drawn, unaged.
   (3) Condition AU: Annealed (hot finished or cold drawn), unaged.

b. Form 2 – Bar, Rod, Age Hardened (for bar and rod formed by forging, see 1.2.1 and form 7).
   (1) Condition HA: Hot finished and age hardened.
   (2) Condition CA: Cold drawn and age hardened.
   (3) Condition AA: Annealed (hot finished or cold drawn) and age hardened.

c. Form 3 – Sheet.
   (1) Condition AU: Cold rolled and annealed.
   (2) Condition AA: Cold rolled, annealed, and age hardened.

d. Form 4 – Strip.
   (1) Condition AU: Cold rolled and annealed.
   (2) Condition AA: Cold rolled, annealed, and age hardened.
   (3) Condition KU: Cold rolled, half hard, as rolled.
   (4) Condition CU: Cold rolled, full hard, as rolled.
   (5) Condition KA: Cold rolled, half hard, and age hardened.
   (6) Condition CA: Cold rolled, full hard, and age hardened.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the accuracy of the address information using the ASSIST online database at https://assist.dla.mil.

FSC 9525

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

e. Form 5 – Wire.
   (1) Condition CU: Cold drawn, as drawn.
   (2) Condition AU: Cold drawn and annealed.
   (3) Condition SU: Cold drawn, spring temper, as drawn.
   (4) Condition AA: Cold drawn, annealed, and age hardened.
   (5) Condition CA: Cold drawn (as drawn) and age hardened.
   (6) Condition SA: Cold drawn, spring temper, and age hardened.

f. Form 6 – Plate.
   (1) Condition AU: Hot rolled and annealed.
   (2) Condition AA: Hot rolled, annealed, and age hardened.

g. Form 7 – Forgings (including bar and rod formed by forging).
   (1) Condition HU: Hot finished, unaged.
   (2) Condition AU: Annealed, unaged.
   (3) Condition HA: Hot finished and age hardened.
   (4) Condition AA: Annealed and age hardened.

1.2.1 Constant cross-section bars and rods formed by forging.

1.2.1.1 Forms 1 or 2. Constant cross-section bars and rods formed by forging are considered to be form 1 or 2 only for purposes of:

a. Chemical analysis (see 4.2.2.1).
b. Slow strain rate tensile tests (see 4.2.2.2).
c. Liquid penetrant testing (see 4.2.2.8).
d. Grain size determination (see 4.2.2.9).
e. Product uniformity testing (see 4.2.2.10).

1.2.1.2 Form 7. Constant cross-section bars and rods formed by forging are considered to be form 7 only for purposes of:

a. Sample frequency for mechanical properties (see 4.2.2.3).
b. Visual and dimensional inspections (see 4.2.2.6).
c. Ultrasonic testing requirements (see 4.2.2.7).

1.2.2 Finishes. The material shall be furnished in the following finishes, as specified (see 6.2 and 6.6.1).

a. Finish A: Hot finished.
b. Finish B: Hot finished forging stock.
c. Finish C: As drawn.
d. Finish D: Rough turned or rough ground (rounds only).
e. Finish E: Semi-smooth machined (rounds only).
f. Finish F: Smooth finished machined (rounds only).
g. Finish G: Bright finish (intended for shafting applications).
h. Finish H: Pickled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

MILITARY STANDARDS

MIL-STD-1684 (controlled distribution) - Control of Heat Treatment

MIL-STD-2132 (controlled distribution) - Nondestructive Examination Requirements for Special Applications

(Copies of MIL-STD-1684 and MIL-STD-2132 may be requested from CommandStandards@navy.mil addressed with a need-to-know.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM INTERNATIONAL

ASTM B880 - Standard Specification for General Requirements for Chemical Check Analysis Limits for Nickel, Nickel Alloys and Cobalt Alloys

ASTM E8/E8M - Standard Test Methods for Tension Testing of Metallic Materials

ASTM E10 - Standard Test Method for Brinell Hardness of Metallic Materials

ASTM E18 - Standard Test Method for Rockwell Hardness of Metallic Materials

ASTM E55 - Standard Practice for Sampling Wrought Nonferrous Metals and Alloys for Determination of Chemical Composition

ASTM E88 - Standard Practice for Sampling Nonferrous Metals and Alloys in Cast Form for Determination of Chemical Composition

ASTM E112 - Standard Test Methods for Determining Average Grain Size

ASTM E290 - Standard Test Method for Bend Testing of Material for Ductility

ASTM E340 - Standard Test Method for Macroetching Metals and Alloys

ASTM E1473 - Standard Test Methods for Chemical Analysis of Nickel, Cobalt, and High-Temperature Alloys

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

AMS 2750 - Pyrometry

AMS 2806 - Identification Bars, Wire, Mechanical Tubing, and Extrusions, Carbon and Alloy Steels and Corrosion and Heat-Resistant Steels and Alloys

AMS 2807 - Identification, Carbon and Low-Alloy Steels, Corrosion and Heat-Resistant Steels and Alloys Sheet, Strip, Plate, and Aircraft Tubing

(Copies of these documents are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.)

3. REQUIREMENTS

3.1 Processing. Nickel-copper-aluminum alloy shall be produced by the electric arc furnace or induction furnace process and further refined by any process proved adequate to provide material that will pass the slow strain rate tensile test requirements (see 3.4 and 4.3.6). Casting may be by ingot or continuous cast methods. When the outputs from multiple heats are used to produce a single continuous cast ingot, the ingot shall be cut to separate the heats and the transition material shall be discarded. Cast ingots or billets shall be hot worked to a fine uniform grain structure. Final processing shall be by hot working or cold working, and heat treating.

3.1.1 Forgings as final product. For all forgings other than bar or rod, a forging sketch shall be prepared when specified (see 6.2). Forgings with final forged thickness of 4 inches or greater, where thickness is defined as the diameter of the greatest inscribed sphere at any location in the forging, shall be to such a configuration that they can be ultrasonically inspected 100 percent by volume with three-directional coverage.

3.1.2 Heat treatment. Heat treatments shall be controlled in accordance with AMS 2750 as modified herein. When specified, MIL-STD-1684, as modified herein, may be used in lieu of AMS 2750 (see 6.2). Furnaces used to solution anneal or age shall have an established working zone with a temperature uniformity range of ±25 °F or less (furnace classes 1 to 5) for the entire range of temperatures used. Load sensors (contact thermocouples) shall not be used to allow heat treatment outside an established work zone or at temperatures above or below the range of temperatures established.

3.1.2.1 Solution annealing. Solution annealing shall be accomplished by holding at a minimum temperature of 1700 °F, followed by water quenching (see 6.8.1.2). The time at temperature must be such that the entire volume of material is heated to the solution annealing temperature.

3.1.2.2 Age hardening. Age hardening shall be accomplished by any process proved adequate to meet the mechanical property requirements in tables I, II, III, and IV, as applicable (see 6.8.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE I. Mechanical properties of bar, rod, forged parts, and plate, age hardened.

Form	Condition	Diameter (D) or maximum distance between parallel surfaces (inches)	Ultimate Tensile strength (lb/in^2)	Yield strength 0.2% offset (lb/in^2, minimum)	Elongation in 2 inches or 4D (percent, minimum) 1/
Rounds	HA, Hot finished and age hardened	Up to 4¼, incl	140,000 min	100,000	20.0
		Over 4¼	140,000 min	100,000	17.0
	AA, Annealed (hot finished or cold drawn) and age hardened	Up to 1	130,000 to 180,000	90,000	20.0
		1 and over	130,000 to 180,000	85,000	20.0
Hexagons, squares, rectangles, and forged parts 2/	HA, Hot finished and age hardened	All sizes	140,000 min	100,000	20.0
	AA, Annealed (hot finished or cold drawn) and age hardened	Up to 1	130,000 to 180,000	90,000	20.0
		1 and over	130,000 to 180,000	85,000	20.0
Rounds	CA, Cold drawn and age hardened	¼ to 1, incl	145,000 min	110,000	15.0
		Over 1 to 3, incl	140,000 min	100,000	17.0
		Over 3 to 4, incl	135,000 min	95,000	20.0
Hexagons	CA, Cold drawn and age hardened	¼ to 2, incl	140,000 min	100,000	15.0
Plate	AA, Hot rolled, annealed, and age hardened	Up to 4, incl	130,000 to 180,000	80,000	20.0

NOTES:

1/ Not applicable to subsize tensile specimens less than 0.250 inch diameter (see 4.3.2.3).

2/ See 3.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE II. Mechanical properties of sheet and strip.

Form and condition		Tensile strength (lb/in², minimum)	Yield strength at 0.2% offset (lb/in², minimum) 1/	Elongation in 2 inches (percent, minimum) 1/	Rockwell hardness
Sheet	AU, Cold rolled and annealed	--	--	--	B85 max
	AA, Cold rolled, annealed, and age hardened	130,000	90,000	15	--
Strip	AU, Cold rolled and annealed	--	--	--	B85 max
	CU, Cold rolled, annealed, and age hardened	130,000	90,000	15	--
	KU, Cold rolled, half hard, as rolled	--	--	--	C20 min
	KA, Cold rolled, half hard, and age hardened	145,000	110,000	8	--
	CU, Cold rolled, full hard, as rolled	--	--	--	C25 min
	CA, Cold rolled, full hard, and age hardened	170,000	130,000	5	--

NOTE:

1/ No yield strength or elongation requirements for sheet and strip under 0.020 inch thick.

TABLE III. Tensile strength of cold drawn wire in coils.

Condition	Size (inches)	Tensile strength (lb/in²)
CU, Cold drawn, as drawn	All sizes	110,000 to 155,000
AU, Cold drawn and annealed	All sizes	110,000 max
SU, Cold drawn, spring temper, as drawn	0.057 and less	165,000 min 1/
	Over 0.057 to 0.114, incl	155,000 min 1/
	Over 0.114 to 0.229, incl	150,000 min 1/
	Over 0.229 to 0.312, incl	145,000 min 1/
	Over 0.312 to 0.375, incl	135,000 min 1/
	Over 0.375 to 0.437, incl	125,000 min
	Over 0.437 to 0.563, incl	120,000 min
AA, Cold drawn, annealed, and age hardened	All sizes	130,000 min
CA, Cold drawn, as drawn, age hardened	All sizes	155,000 min
SA, Cold drawn, spring temper, and age hardened	Up to 0.114, incl	180,000 min
	Over 0.114 to 0.375, incl	170,000 min
	Over 0.375 to 0.563, incl	160,000 min

NOTE:

1/ Applicable to material in coil. For material in straightened and cut lengths, deduct 15,000 lb/in² from above values.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE IV. Bending properties of cold rolled sheet and strip. [1/]

Thickness of sheet and strip (inch)		Radius of pin or mandrel, multiples of the thickness of the material	Angle of bend (degrees)		
From	To (incl)		Condition AU, cold rolled and annealed sheet and strip	Condition KU, cold rolled, half hard, as rolled strip	Condition AA, cold rolled, annealed and age hardened, sheet and strip, and CU, cold rolled, full hard, as rolled strip
Up to 0.010	0.031	0.5	180	120	90
Over 0.031	0.062	0.5	180	120	90
Over 0.062	0.125	0.5	180	120	90
Over 0.125	0.250	1.0	180	120	90

NOTE:

[1/] There are no bending requirements for condition KA, cold rolled, half hard aged or condition CA, cold rolled, full hard aged strip.

3.2 Chemical requirements. Chemical composition of the material shall be in accordance with table V. When a product uniformity test is required (see 6.2), the material shall be in accordance with the requirements of table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE V. Chemical composition.

Element	Requirement (percent) (maximum unless a range or minimum is indicated)
Nickel 1/	63.0 minimum
Aluminum	2.50 to 3.15
Carbon (total)	0.18
Iron	2.0
Manganese	1.5
Silicon	0.50
Titanium	0.35 to 0.85
Cobalt	0.25
Sulfur	0.006
Copper 2/	27.0 to 33.0
Tin	0.006
Phosphorus	0.02
Zinc	0.01
Lead	0.006

NOTES:

1/ Element may be determined arithmetically by difference.

2/ Copper content, or combined silver and copper content when the analysis method does not differentiate between copper and silver. A separate silver analysis is not required.

3.3 Mechanical properties. Mechanical properties shall be as specified in tables I, II, III, IV, and VI as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE VI. Mechanical properties of bar, rod, forged parts, and plate, unaged. [1/]

Form	Condition	Size (inches)	Hardness (maximum)	
			Brinell (3000 kg)	Rockwell
Rounds	HU, Hot finished, unaged	Up to 4¼, incl	245	C23
		Over 4¼	260	C26
	AU, Annealed (hot finished or cold drawn), unaged	All	185	B90
Hexagons, squares, rectangles, and forged parts	HU, Hot finished, unaged	All	245	C23
	AU, Annealed (hot finished or cold drawn), unaged	All	185	B90
Hexagons	CU, Cold drawn, unaged	All	260	C26
Rounds	CU, Cold drawn, unaged	¼ to 1, incl	280	C29
		Over 1 to 3, incl	260	C26
		Over 3 to 4, incl	240	C22
Plate	AU, Hot rolled and annealed	Up to 4, incl	185	B90

NOTE:

[1/] No tensile tests are required except as provided for in 4.3.2.2.

3.3.1 Unaged material. Material ordered in the unaged condition shall, after proper heat treatment, be capable of meeting the minimum mechanical properties of the corresponding age hardened condition of the forms specified in tables I, II, III, and IV, as applicable.

3.3.2 Bending properties (forms 3 and 4). Sheet and strip, except half hard and age hardened or full hard and age hardened strip, shall not crack when subjected to cold bending (see 4.3.2.5).

3.3.3 Forged rings and discs. When specified (see 6.2), forged rings and discs shall have a minimum hardness of 24 Rockwell C hardness number (HRC) (265 Brinell hardness number [BHN]) in lieu of the mechanical properties specified in table I.

3.4 Susceptibility to intergranular cracking. The material shall not be susceptible to intergranular cracking (see 4.3.6).

3.5 Grain size of bar and rod.

3.5.1 Unaged bar and rod (form 1). When specified (see 6.2), the ASTM grain size number of unaged bar and rod with a diameter or minimum distance between parallel surfaces of 3½ inches or less shall be 5 or finer. When specified (see 6.2), the ASTM grain size number of unaged bar and rod with a diameter or minimum distance between parallel surfaces of over 3½ inches shall be 3 or finer.

3.5.2 Age hardened bar and rod (form 2). The ASTM grain size number of age hardened bar and rod with a diameter or minimum distance between parallel surfaces of 3½ inches or less shall be 5 or finer. The ASTM grain size number of bar and rod with a diameter or minimum distance between parallel surfaces of over 3½ inches shall be 3 or finer.

3.6 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

3.7 Soundness. Material shall be of uniform quality and condition, free of defects harmful to its intended use, such as seams, pipe, cracks, excessive scale, fins, porosity, and segregation, as determined by visual examination and nondestructive testing.

3.7.1 Ultrasonic inspection.

3.7.1.1 Ultrasonic inspection of large size bars, rods, and forgings (forms 2 and 7). Bars, rods, and forgings with a thickness 4 inches or greater, where thickness is defined as the diameter of the greatest inscribed sphere at any location in the bar or forging, shall be ultrasonically inspected as specified in 4.3.3.1 and 4.3.3.2. Inspection shall be performed after the aging treatment. Ultrasonic procedures other than those specified may be employed but they shall be approved by the purchaser.

3.7.1.2 Additional, elective ultrasonic inspection of bars, rods, plates, and forgings (forms 1, 2, 6, and 7). When specified (see 6.2), bars, rods, plate, and forgings shall be ultrasonically inspected as specified in 4.3.3.1 and 4.3.3.3. Ultrasonic inspection, in accordance with this paragraph, does not fulfill the ultrasonic inspection requirement in 3.7.1.1.

3.7.2 Liquid penetrant inspection. When specified (see 6.2), bar, rod, plate, and forgings shall be liquid penetrant inspected as specified in 4.3.4.

3.8 Wrapping requirements (form 5). Wire, ¼ inch diameter or less, in condition CU (cold drawn, as drawn), condition AU (cold drawn and annealed), or condition SU (cold drawn, spring temper, as drawn) shall be capable of being wrapped around a rod of the same diameter as the wire without cracking. There are no wrapping requirements for wire greater than ¼ inch in diameter. There are no wrapping requirements for wire in conditions AA (cold drawn, annealed, and age hardened), CA (cold drawn [as drawn] and age hardened), and SA (cold drawn, spring temper, and age hardened).

3.9 Dimensional requirements. Unless otherwise specified (see 6.2), the dimensional tolerances shall be as specified herein.

3.9.1 Bar and rod.

3.9.1.1 Diameter, distance between parallel faces, or thickness. The diameter, distance between parallel faces, or thickness of hot finished or cold drawn bar and rod, aged and unaged, shall not vary at any point by more than the amounts specified in tables VII or VIII, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE VII. Tolerances on diameter or distance between parallel faces of conditions HU (hot finished, unaged) and HA (hot finished, age hardened), bar and rod.

Nominal diameter or distance between parallel faces	Size (inches)	Tolerances (inches) 1/, 2/	
		Plus	Minus
Rod and bar, finish A, hot finished	1 and under	0.016	0.016
	Over 1 to 2, incl	0.031	0.016
	Over 2 to 4, incl	0.047	0.031
	Over 4	0.125	0.063
Round rod, finish D, rough turned or rough ground	1 and over	0.031	0.010
Round rod, finish E, semi-smooth, machined	Over 3½	0.031	0.000
Round rod, finish F, hot finished, smooth finish, machined	Over 3½	0.000	0.005 3/
Round rod, finish B, hot finished forging stock, finish	Under 1	0.005	0.010 4/
	1 and over	0.031	0.010 4/

NOTES:

1/ Tolerances apply to diameter of rounds to distance between parallel surfaces for hexagons and squares and separately to width and thickness of rectangles.

2/ Areas on the ends of the bar and rod (for approximately 1 inch) that are under the permissible variations in diameter are permitted provided the desired finished surface can be obtained within the machining allowance (see 3.11).

3/ Permissible variations available as +0.005 inch, -0 inch; or +0.0025 inch, -0.0025 inch, when specified (see 6.2).

4/ Spot grinding may be permitted to remove minor surface defects. The depth of grinding shall not exceed 3 percent of the diameter.

TABLE VIII. Tolerances on diameter or distance between parallel faces of finish B (hot finished forging stock) as drawn, rod and bar.

Nominal diameter or distance between parallel faces (inches)		Tolerances (inch)	
		Plus	Minus
Rounds	Under ½	0	0.003
	½ to $^{15}/_{16}$, incl	0	0.002
	Over $^{15}/_{16}$ to $1^{15}/_{16}$, incl	0	0.003
	Over $1^{15}/_{16}$ to 2½, incl	0	0.004
	Over 2½ to 3, incl	0	0.005
	Over 3 to 3½, incl	0	0.006
	Over 3½ to 4, incl	0	0.007
Hexagons	Up to 2, incl	0	0.009

3.9.1.2 Out of roundness. Round rod, all sizes and in straight lengths, shall not be out of round by more than one-half the permissible tolerances in diameter specified in tables VII and VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

3.9.1.3 Length.

3.9.1.3.1 Bar and rod. Unless otherwise specified (see 6.2), bar and rod shall be furnished in random mill lengths with cut or sheared ends. Rod and bar furnished to nominal (stock) lengths shall be furnished with either cropped or cut ends. Material ordered to cut lengths shall be furnished with square cut or machined ends. Where rod and bar is ordered in multiples of a specified unit length, ¼ inch shall be allowed for each multiple cut unless otherwise specified (see 6.2), and the rod and bar shall have cut ends.

3.9.1.3.2 Permissible variations in length. The permissible variations in lengths of rod and bar, all conditions, shall be as specified in table IX.

TABLE IX. Permissible variations in length of bar and rod, all conditions.

Specified length (feet)		Shortest acceptable lengths (feet)	Maximum acceptable length	Maximum permissible percentage by weight of short length
Random mill lengths [1/]		6	24 feet	25% under 9 feet
Nominal (stock) length [2/]	16 to 18	16	18 ft., ½ inch	100% 16 feet or longer
	14 to 16	14	16 ft., ½ inch	100% 14 feet or longer
	12 to 14	12	14 ft., ½ inch	100% 12 feet or longer
	10 to 12 [3/]	10	12 ft., ½ inch	100% 10 feet or longer
	8 to 10 [3/]	8	10 ft., ½ inch	100% 8 feet or longer
	6 to 8 [3/]	6	8 ft., ½ inch	100% 6 feet or longer
	4 to 6	4	6 ft., ½ inch	100% 4 feet or longer
Cut-to length [4/]		Specified length	Specified length plus ⅛ inch	100% specified plus ⅛ inch

NOTES:

1/ For hot finished sections weighing over 25 pounds per lineal foot and for smooth forged products, all sections, short lengths down to 2 feet may be furnished.

2/ Other nominal lengths with a specified range of not less than 2 feet, with no shorts allowed, may also be furnished.

3/ For cold drawn rod and bar under ½ inch in diameter or distance across flats ordered to nominal or stock lengths with a 2-foot range, at least 93 percent of such material shall be within the range specified; the balance may be in shorter lengths but in no case shall lengths less than 4 feet be furnished.

4/ For diameters over 8 inches, the tolerance shall be +¼ inch, -0 inch.

3.9.1.4 Edges. Unless otherwise specified (see 6.2), square, rectangular, and hexagonal bar and rod shall have angles and corners consistent with commercial practice.

3.9.1.5 Straightness. The permissible variations in straightness of rod and bar, as determined by the departure from the true straightness (throw in one revolution or depth of chord [see 4.2.3.1 and 4.2.3.2]), shall be as specified in table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE X. Permissible variations in straightness of rod and bar. [1/]

Condition	Finish and size	Permissible deviations in straightness (inch)
Condition AA and AU hot finished	Bar and rod: Finish A, hot finished [2/]	Depth of chord [3/] 0.050 per foot of length
	Rounds: Finish D, rough turned or rough ground	Throw in one revolution [4/] or depth of chord [3/] 0.050 per foot of length
	Rounds: Finish E, semi-smooth machined Finish F, smooth finished machined	Throw in one revolution [4/] 0.0031 per foot of length 0.0015 per foot of length
Condition CA and CU cold drawn	Rounds (diameter): Finish C, as drawn Up to 4 inches, incl	Depth of chord [3/] 0.030 per foot of length
	Hexagons and squares: Finish C, as drawn All sizes	Depth of chord [3/] 0.030 per foot of length

NOTES:

1/ Not applicable to finish B, hot finished forging stock.

2/ Except forging quality stock.

3/ The maximum curvature (depth of chord) shall not exceed the value indicated multiplied by length in feet.

4/ The throw in one revolution in any 20 feet maximum length shall not exceed the values indicated multiplied by length in feet.

3.9.2 Forged parts dimensions. All forgings shall conform to the sizes and shapes as specified (see 6.2).

3.9.3 Plate, sheet, and strip.

3.9.3.1 Thickness. The thickness tolerances of cold rolled sheet and strip, all conditions, shall conform to the requirements specified in table XI. The tolerances on thickness of hot rolled plate, all conditions, shall conform to the requirements specified in table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XI. Tolerances on thickness of cold rolled sheet and strip, all conditions.

Nominal thickness (inches)		Tolerances on thickness for widths given [1/] (inch ±)		
		Less than 12 inches wide [1/]	48 inches wide or less	Over 48 to 60 inches wide, incl
Sheet	0.018 to 0.025, incl	--	0.002	0.003
	Over 0.025 to 0.034, incl	--	0.003	0.004
	Over 0.034 to 0.043, incl	--	0.004	0.005
	Over 0.043 to 0.056, incl	--	0.004	0.005
	Over 0.056 to 0.070, incl	--	0.005	0.006
	Over 0.070 to 0.078, incl	--	0.006	0.007
	Over 0.078 to 0.093, incl	--	0.007	0.008
	Over 0.093 to 0.109, incl	--	0.007	0.009
	Over 0.109 to 0.125, incl	--	0.008	0.010
	Over 0.125 to 0.140, incl	--	0.008	0.010
	Over 0.140 to 0.171, incl	--	0.009	0.012
	Over 0.171 to 0.187, incl	--	0.010	0.013
	Over 0.187 to 0.218, incl	--	0.011	0.015
	Over 0.218 to 0.234, incl	--	0.012	0.016
	Over 0.234 to 0.250, incl	--	0.013	0.018
Strip	0.010 to 0.050, incl	0.0015	--	--
	Over 0.050 to 0.093, incl	0.0025	--	--
	Over 0.093 to 0.125, incl	0.0040	--	--
	Over 0.125 to 0.156, incl [2/]	0.0045 [3/]	--	--
	Over 0.156 to 0.250, incl [2/]	0.0055 [3/]	--	--

NOTES:

[1/] Measured ⅜ inch or more from any edge except for strip under 1 inch in width, which is measured at any place.

[2/] Available in straightened and cut lengths only.

[3/] For widths over 8 inches, the permissible variations for cold rolled sheet are applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XII. Permissible variations in weight and thickness of hot rolled rectangular plate. 1/, 2/

Specified thickness 3/ (inches)	Permissible excess in average weight per square foot of plate for widths given in inches expressed in percentage of nominal weights 4/									
	Under 48	48 to 60, excl	60 to 72, excl	72 to 84, excl	84 to 96, excl	96 to 108, excl	108 to 120, excl	120 to 132, excl	132 to 144, excl	144 to 160, excl
3/16 to 5/16, excl	9.0	10.5	12.0	13.5	15.0	16.5	18.0	--	--	--
5/16 to 3/8, excl	7.5	9.0	10.5	12.0	13.5	15.0	16.5	18.0	--	--
3/8 to 7/16, excl	7.0	7.5	9.0	10.5	12.0	13.5	15.0	16.5	18.0	--
7/16 to 1/2, excl	6.0	7.0	7.5	9.0	10.5	12.0	13.5	15.0	16.5	18.0
1/2 to 5/8, excl	5.0	6.0	7.0	7.5	9.0	10.5	12.0	13.5	15.0	16.5
5/8 to 3/4, excl	4.5	5.5	6.0	7.0	7.5	9.0	10.5	12.0	13.5	15.0
3/4 to 1, excl	4.0	4.5	5.5	6.0	7.0	7.5	9.0	10.5	12.0	13.5
1 to 1¼, excl	4.0	4.0	4.5	5.5	6.0	7.0	7.5	9.0	10.5	12.0
1¼ to 1½, excl	4.0	4.0	4.5	5.5	6.0	7.0	7.5	9.0	10.5	--
1½ to 1¾, excl	4.0	4.0	4.5	5.5	6.0	7.0	7.5	9.0	--	--
1¾ to 2, incl	4.0	4.0	4.5	5.5	6.0	7.0	7.5	--	--	--

NOTES:

1/ The term "lot" applied to the above table is defined as all of the plate of each group width and each group thickness.

2/ No plates shall vary more than 0.01 inch under the thickness ordered, and the overweight of each lot in each shipment shall not exceed the amount shown in the table. Spot grinding will be permitted to remove surface imperfections; such spots shall not exceed 0.01 inch under the specified thickness.

3/ All plates shall be ordered by thickness and not weight, per square foot.

4/ The weight of individual plates shall not exceed the nominal weight by more than 1¼ times the amount shown.

3.9.3.2 Width. The width of sheet and strip shall not vary from the ordered width by more than the amount specified in table XIII and those for plate shall be as specified in table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XIII. Tolerances on width of cold rolled sheet and strip, all conditions.

Nominal thickness (inch)		Nominal width (inches)	Tolerances on widths (inch)	
			Plus	Minus
Sheet	All thickness	All widths	0.125	0.000
Strip, split edge	0.010 to 0.024, incl	12 and under	0.007	0.007
	Over 0.024 to 0.075, incl	12 and under	0.007	0.007
	Over 0.075 to 0.100, incl	12 and under	0.009	0.009
	Over 0.100 to 0.125, incl	12 and under	0.012	0.012
	Over 0.125 to 0.250, incl 1/	8 and under	0.015	0.015
	Over 0.125 to 0.250, incl 1/, 2/	Over 8 to 12, incl	0.125	0

NOTES:

1/ Available in straightened and cut lengths only.

2/ Slit edge or sheared edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XIV. Permissible variations in width of sheared, machined, abrasive, or plasma-torch-cut rectangular plate. 1/ 2/

Specified thickness (inches)		Permissible variation in width for widths given (inch)									
		Up to 30, incl		Over 30 to 72, incl		Over 72 to 108, incl		Over 108 to 144, incl		Over 144 to 160, incl	
		Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus
Sheared 3/	3/16 to 5/16, excl	3/16	1/8	1/4	1/8	3/8	1/8	1/2	1/8	--	--
	5/16 to 1/2, excl	1/4	1/8	3/8	1/8	3/8	1/8	1/2	1/8	5/8	1/8
	1/2 to 3/4, excl	3/8	1/8	3/8	1/8	1/2	1/8	5/8	1/8	3/4	1/8
	3/4 to 1, excl	1/2	1/8	1/2	1/8	5/8	1/8	3/4	1/8	7/8	1/8
	1 to 1 1/4, incl	5/8	1/8	5/8	1/8	3/4	1/8	7/8	1/8	1	1/8
Machined or abrasive cut 4/ 5/	3/16 to 1 1/4, incl	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
	Over 1 1/4 to 4 incl	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8
Plasma-torch cut 6/	3/16 to 2, excl	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0
	2 to 3, incl	5/8	0	5/8	0	5/8	0	5/8	0	5/8	0

NOTES:

1/ Permissible variations in width for powder cut or inert arc cut plate or machined, powder cut or inert arc cut circular plate shall be as agreed upon between the manufacturer and the purchaser.

2/ Tolerances for plasma-torch-cut sketch plate shall be as agreed upon between the manufacturer and the purchaser.

3/ The minimum sheared width is 10 inches for material 3/4 inch and under in thickness and 20 inches for material over 3/4 inch in thickness.

4/ The minimum abrasive cut width is 2 inches.

5/ Tolerances shown are applicable for lengths up to 240 inches. For lengths over 240 inches, an additional 1/16 inch is permitted, both plus and minus.

6/ Tolerances shown for plasma-torch cutting may be obtained all on the minus side, or divided between the plus and minus side, as specified (see 6.2).

3.9.3.3 Lengths.

3.9.3.3.1 Sheet. Unless otherwise specified (see 6.2), sheet shall be furnished in cut lengths. The permissible variation in length shall be +1/8 inch, -0 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

3.9.3.3.2 Strip. Unless nominal or stock lengths or cut lengths are specified, strip, in conditions other than age hardened, shall be furnished in coil in thicknesses up through 0.125 inch and in random straight lengths in thicknesses over 0.125 inch. Unless otherwise specified (see 6.2), age hardened strip shall be furnished in straightened (cut) lengths only.

a. The length of cut length strip shall not vary under or over the ordered length by more than ⅛ inch.

b. The length of strip up to 12 inches wide, in nominal or stock lengths, shall not vary from the nominal or stock length by more than ½ inch, except for the permissible amounts of short lengths specified in table XV. The maximum percentage by weight of short lengths and the required percentage by weight of nominal or stock lengths in any shipment shall be as specified in table XV.

TABLE XV. Permissible variations in stock lengths of strip, up to 12 inches wide.

Nominal or stock length (feet)	Required percentage by weight of stock length	Maximum permissible percentages, by weight, of short lengths				
		Over 8 to 10 feet, incl	Over 6 to 8 feet, incl	Over 4 to 6 feet, incl	Over 2 to 4 feet, incl	Under 2 feet
10	60	40	30	20	10	0
8	70	--	30	20	10	0
6	80	--	--	20	10	0

3.9.3.3.3 Plate. The permissible variations in length for plate shall be as specified in table XVI. The permissible variations in diameter for circular plate shall be as specified in table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XVI. Permissible variations in width of sheared, machined, abrasive, or plasma-torch-cut rectangular plate. [1/]

Specified thickness (inches)		Permissible variation in length for lengths given (inch)													
		60 and under		Over 60 to 96, incl		Over 96 to 120, incl		Over 120 to 240, incl		Over 240 to 360, incl		Over 360 to 450, incl		Over 450 to 540, incl	
		Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus	Plus	Minus
Sheared [2/]	3/16 to 5/16, excl	3/16	1/8	1/4	1/8	3/8	1/8	1/2	1/8	5/8	1/8	3/4	1/8	--	--
	5/16 to 1/2, excl	3/8	1/8	1/2	1/8	1/2	1/8	1/2	1/8	5/8	1/8	3/4	1/8	7/8	1/8
	1/2 to 3/4, excl	1/2	1/8	1/2	1/8	5/8	1/8	5/8	1/8	3/4	1/8	7/8	1/8	1 1/8	1/8
	3/4 to 1, excl	5/8	1/8	5/8	1/8	5/8	1/8	3/4	1/8	7/8	1/8	1 1/8	1/8	1 3/8	1/8
	1 to 1 1/4, incl	3/4	1/8	3/4	1/8	3/4	1/8	7/8	1/8	1 1/8	1/8	1 3/8	1/8	1 5/8	1/8
Machined or abrasive cut [3/]	3/16 to 1 1/4, incl	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
	1 1/4 to 4, incl	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8	3/16	1/8
Plasma-torch cut [4/ 5/]	3/16 to 2, incl	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0
	2 to 2 1/4	5/8	0	5/8	0	5/8	0	5/8	0	5/8	0	5/8	0	5/8	0

NOTES:

1/ Permissible variations in length for powder cut or inert arc cut plate or machined, powder cut, or inert arc cut circular plate shall be as agreed upon between the manufacturer and the purchaser.

2/ The minimum sheared length is 10 inches.

3/ Abrasive cut applicable to a maximum length of 144 to 400 inches depending on the thickness and width ordered.

4/ Tolerances for plasma-torch-cut sketch plate shall be as agreed upon between the manufacturer and the purchaser.

5/ Tolerances for plasma-torch cutting may be obtained all on the minus side or divided between the plus or minus side as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XVII. Permissible variation in diameter for circular plate.

Specified diameter (inches)		Thickness (inches, maximum)	Permissible variations over specified diameter for thickness given (inch)					
			Thickness to 3/8, incl		3/16 to 2, excl		2 to maximum thickness	
			Plus	Minus	Plus	Minus	Plus	Minus
Sheared plate	20 to 32, excl	3/8	1/4	0	--	--	--	--
	32 to 84, excl	3/8	5/16	0	--	--	--	--
	84 to 108, excl	3/8	3/8	0	--	--	--	--
	108 to 140, excl	3/8	7/16	0	--	--	--	-
Plasma-torch-cut plate 1/ 2/	19 to 20, excl	3	--	--	1/2	0	5/8	0
	20 to 22, excl	2 3/4	--	--	1/2	0	5/8	0
	22 to 24, excl	2 1/2	--	--	1/2	0	5/8	0
	24 to 28, excl	2 1/4	--	--	1/2	0	5/8	0
	28 to 32, excl	2	--	--	1/2	0	5/8	0
	32 to 34, excl	1 3/4	--	--	1/2	0	--	--
	34 to 38, excl	1 1/2	--	--	1/2	0	--	--
	38 to 40, excl	1 1/4	--	--	1/2	0	--	--
	40 to 140, excl	3	--	--	1/2	0	5/8	0

NOTES:

1/ The tolerance for plasma-torch-cut plate may be obtained all on the minus side or divided between the plus and minus sides, as specified (see 6.2).

2/ Permissible variations in plasma-torch-cut sketch plate shall be as agreed upon between the manufacturer and the purchaser.

3.9.3.4 Edges.

3.9.3.4.1 Sheet. Unless otherwise specified (see 6.2), sheet shall be furnished with edges as would result from slitting, sawing, or shearing.

3.9.3.4.2 Strip. Unless otherwise specified (see 6.2), strip shall be furnished with the type of edge as specified in table XVIII.

TABLE XVIII. Type of edge of strip.

Thickness (inches)	Width (inches)	Type of edge
0.125 and less	12 and less	Slit 1/
Over 0.125	8 and less	Slit 2/
Over 0.125	Over 8	Slit or sheared 2/

NOTES:

1/ In coil or straight (random or cut) lengths.

2/ In random or cut straight lengths only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

3.9.3.4.3 Plate. Unless otherwise specified (see 6.2), plate 1¼ inch in thickness and less shall be furnished with sheared edges, and plate over 1¼ inch in thickness shall be furnished with machined, abrasive, laser, or plasma-torch-cut edges.

3.9.3.5 Straightness. The edgewise curvature or maximum departure from straight line in any longitudinal edge on sheet and strip cut to length or in random, nominal, or stock lengths shall not exceed 0.05 inch per foot of length.

3.9.4 Wire.

3.9.4.1 Diameter. The permissible tolerances in diameter of all conditions of wire shall be as specified in table XIX.

TABLE XIX. Tolerances in diameter of cold drawn wire, all conditions.

Specified diameter(inch)	Tolerances (inch)	
	Plus	Minus
To 0.0044, incl	0.0002	0.0002
Over 0.0044 to 0.0079, incl	0.00025	0.00025
Over 0.0079 to 0.0149, incl	0.0003	0.0003
Over 0.0149 to 0.0199, incl	0.0004	0.0004
Over 0.0199 to 0.031, incl	0.0005	0.0005
Over 0.031 to 0.045, incl	0.0006	0.0006
Over 0.045 to 0.079, incl	0.0007	0.0007
Over 0.079 to 0.1875, incl	0.001	0.001
Over 0.1875 to 0.406, incl	0.001	0.002
Over 0.406	0.002	0.002

3.9.4.2 Out of roundness. Wire shall not be out of round by more than one-half the total permissible tolerances specified in table XIX.

3.9.4.3 Coil and spool weights. Permissible variations in coil and spool weights of cold drawn wire shall be as stated in tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XX. Permissible variation in weight of cold drawn wire in coils and on spools (except Air Force use).

Condition	Wire diameter (inch)	Standard spool weight (pounds)	Maximum weight wire on each spool (pounds)
Spools (all conditions)	Under 0.010	2	2½
	0.010 to 0.018, incl	5	6
	Over 0.018 to 0.040, incl	10	15
Condition	**Wire diameter (inch)**	**Approximate mean coil diameter (inches)**	**Maximum weight per coil (pounds)**
Coils (all conditions)	Under 0.010	8	15
	0.010 to 0.018, incl	8	25
	Over 0.018 to 0.040, incl	8 to 12	40
	Over 0.040 to 0.081, incl	16 to 20	100
	Over 0.081 to 0.312, incl	18 to 22	200
	Over 0.312 to 0.563, incl	22 to 30	200

TABLE XXI. Permissible variation in weight of cold drawn wire in coils and on spools (Air Force use only).

Condition	Wire diameter (inch)	Core diameter (inches)	Maximum weight wire on each spool (pounds)
All conditions	Under 0.035	1	1
	0.035 to 0.063, incl	4 to 5	1
	0.064 to 0.0915, incl	--	5
	0.0916 to 0.162, incl	--	25 1/

NOTE:

1/ Coil.

3.10 Marking. Unless otherwise specified (see 6.2), bar, rod, and wire shall be marked in accordance with AMS 2806 and sheet strip and plate shall be marked in accordance with AMS 2807. As a minimum, the marking content shall include this specification number and revision, the form, the condition, the producer's name or trademark, heat number, and lot number. Individual forgings shall be marked with the specification number, condition, the producer's name or trademark, heat number, and lot number. Identification number and markings shall not be made by chisel or other sharp instruments.

3.11 Machining allowance for finish A, hot finished material. Machining allowances shall be made for hot finished material as specified in table XXII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XXII. Machining allowances for hot finished rod and bar (allow the following oversizes).

Condition	Dimensions (inches) 1/	Machining allowances (inch) 2/			
		Diameter	Across flats (hexagons and squares)	Thickness (rectangles)	Width (rectangles)
Finish A, Hot finished	Up to 0.875, incl	0.125	0.125	0.125	0.187
	Over 0.875 to 1.875, incl	0.125	0.187	0.125	0.187
	Over 1.875 to 2.875, incl	0.187	0.250	--	0.187
	Over 2.875 to 3.812, incl	0.250	--	--	0.187
	Over 3.812	0.250	--	--	0.375
Finish D Rough turned or rough ground 2/	0.500 to 0.875, incl	0.005	--	--	--
	Over 0.875 to 4.000, incl	0.062	--	--	--
	Over 4.000 to 12.000, incl	0.125	--	--	--
	Over 12.000 to 20.000, incl	0.125	--	--	--
Finish E Semi-smooth machined	Up to 2.500, incl (Up to 10-ft lengths, incl)	0.062	--	--	--
	Over 2.500 to 4.000, incl (Over 10- to 20-ft lengths, incl)	0.125	--	--	--
	Over 4.000 to 10.000, incl (up to 30-ft lengths, incl)	0.125	--	--	--

NOTES:

1/ Dimensions apply to diameter of rods, distance between parallel surfaces of squares and hexagons, and separately to width or thickness of rectangles.

2/ The allowances are recommended for rounds to be machined in lengths 36 inches or less (such as stepdown shafts) and for squares, hexagons, and rectangles to be machined in lengths 24 inches or less. Lengths longer than those cited should show the finish, cross-sectional dimensions, and also the length in which the material is to be machined in order that material may be supplied with sufficient oversize, including allowance for out-of-straightness.

3.12 Workmanship. The material shall be uniform in quality and condition, clean, smooth, flat, commercially straight, and be free from foreign material and imperfections such as pipe, laps, cracks, and seams which, due to their nature, degree, or extent, may detrimentally affect the suitability for the service intended. Products ordered with a hot finished, rough turned, or semi-smooth machined surface may contain surface imperfections that shall not be considered injurious unless they exceed the recommended machining allowance specified in table XXII. Inspection procedures for hot finished or machined material shall be as specified in 4.2.3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products supplied to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, either indicated or actual, but it does not commit the Government to accept defective material.

4.2 Quality conformance inspection.

4.2.1 Definition of lot.

4.2.1.1 Visual and dimensional inspection, grain size, and mechanical tests. A lot shall consist of material of one heat, form, condition, finish (for visual only), and size, and heat treated in the same furnace charge when ordered in the heat treated condition.

a. Heat treated conditions shall include age hardened, annealed and age hardened, and annealed only.
b. For grain size and mechanical tests, size shall refer to the same thickness of sheet, strip, or plate, and the same cross-section of other products.

4.2.1.2 Chemical analysis and product uniformity test. A lot shall consist of material from one heat that has been refined in a single batch (see 3.1).

4.2.1.3 Slow strain rate tensile tests. A lot shall consist of material from one heat (see 4.2.1.2) that has been heat treated in accordance with the same or equivalent heat treating procedure (see 4.3.6.1.1).

4.2.2 Sampling of lots.

4.2.2.1 Chemical composition (see 4.2.1.2). One representative sample shall be obtained from each lot.

a. When the metal is refined in one batch, the ladle analysis may be used (see 3.1).
b. When the lot is sampled from material in the cast form, the lot shall be sampled in accordance with ASTM E88.
c. When the lot is sampled from wrought material, the lot shall be sampled in accordance with ASTM E55.

4.2.2.2 Slow strain rate tensile tests. Three specimens shall be prepared and tested per lot. When the material is in the aged condition, specimens shall be taken after the final heat treatment. When the material is in the solution annealed condition, specimens shall be taken after the final anneal and shall be aged in accordance with 4.3.6.1. When the material has not been solution annealed, specimens shall be solution annealed and aged in accordance with 4.3.6.1. Specimens that have not received a minimum of a 4 to 1 (75 percent) reduction from the original ingot size shall be hot worked to give them a total reduction of at least 4 to 1. Tests accomplished on one form of material do not need to be repeated for a different size or form from the same lot that has undergone an equivalent heat treatment as defined in 4.3.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.2.2.2.1 Bar and rod. Slow strain rate tensile test specimens shall be taken from one end of bars and rods at the quarter-diameter (mid-radius) for round bar or quarter-thickness for hex, square, or rectangular bar, as applicable, and in the longitudinal direction. Bars and rods too small to have a slow strain rate test specimen taken from the quarter-diameter or quarter-thickness that are taken from a lot (see 4.2.1.3) which does not have any larger sizes shall have the specimens taken from the center and in the longitudinal direction. Bars and rods too small to have a slow strain rate test specimen taken from the center which are taken from a lot (see 4.2.1.3) that does not have any larger sizes shall have the specimens taken from the heat at the latest intermediate rolling step that a slow strain rate test specimen can be taken from the quarter-diameter (mid-radius) or quarter thickness.

4.2.2.2.2 Forgings. Slow strain rate tensile test specimens shall be taken from prolongations, extra forgings, or, if the shape of the finished part permits, removed, without loss, from the forging. All specimens shall be taken in the longitudinal direction. Specimens shall be taken at the quarter-diameter (mid-radius) or at the quarter-thickness. Forgings too small to have a slow strain rate test specimen taken from the quarter-diameter or at the quarter-thickness may have the specimens taken from the center. Forgings too small to have a slow strain rate test specimen taken from the center may be taken from the heat at the latest intermediate forging step that a slow strain rate test specimen can be taken from the mid-radius or quarter-thickness.

4.2.2.2.3 Wire. Slow strain rate tensile tests shall be taken from the heat at the latest intermediate rolling or forging step that a slow strain rate test specimen can be taken from the mid radius and heat treated in accordance with 4.3.6.1.

4.2.2.2.4 Sheet and plate. Slow strain rate tensile tests specimens shall be taken from one end of a plate at the quarter-thickness. Plate too small to have a slow strain rate test specimen taken from the quarter-thickness may have the specimens taken from the center. Sheet and plate too small to have a slow strain rate test specimen taken from the center may have the specimens taken at the latest intermediate rolling step that a slow strain rate test specimen can be taken from the quarter-thickness.

4.2.2.3 Mechanical, bending, and wire wrapping tests for rod, bar, wire, and flat products. One test specimen shall be taken from each lot of 30,000 pounds or less. Two test specimens shall be taken from lots weighing over 30,000 pounds. Specimens shall be removed after final heat treatment.

4.2.2.4 Sampling of lots for mechanical tests of forged parts, 250 pounds and less. Specimens from the lot shall be taken from prolongations unless the shape of the finished parts permit suitable specimens to be removed without loss of the forging, or the specimens may be taken from extra forgings. One specimen shall be taken from lots of 30,000 pounds or less and two specimens shall be taken from lots over 30,000 pounds.

4.2.2.5 Sampling of lots for mechanical tests of forged parts, greater than 250 pounds. Each forging shall be tested. Specimens shall be taken from prolongations unless the shape of the finished part permits a suitable sample piece to be removed without loss of the forging.

4.2.2.6 Sampling for visual and dimensional examination.

4.2.2.6.1 All forms except forgings weighing 250 pounds or more. From each lot, representative samples for inspection of visual and dimensional characteristics including packaging, packing, and marking requirements shall be taken in accordance with table XXIII. In case of wire, a unit shall be one coil of wire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Sample size for visual and dimensional inspections.

Lot size	Sample size
2 to 8	All
9 to 15	8
16 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38

4.2.2.6.2 Forged parts weighing 250 pounds or more. Each forging weighing 250 pounds or more shall be subjected to surface inspection for workmanship and dimensions.

4.2.2.7 Sampling for additional, elective ultrasonic tests. When required (see 3.7.1.2), each piece of each lot shall be inspected 100 percent by volume.

4.2.2.8 Sampling for liquid penetrant tests. When required (see 3.7.2), each piece of each lot shall be inspected.

4.2.2.9 Sampling for grain size determination. When required (see 3.5), a sample for grain size determination shall be taken from a slice at least ⅜ inch thick from one end of a bar, rod, or constant cross-section forging representing each lot.

4.2.2.10 Sampling for product uniformity test. When a product uniformity test is specified (see 3.2 and 6.2), the product shall be sampled in accordance with ASTM E55 or ASTM E88, as applicable, and the requirements herein. Sampling for a product uniformity test may be taken at any of the following steps in the manufacture of the product. Sampling may be done before or after any cropping operations. A separate chemistry shall be determined for each location sampled.

4.2.2.10.1 Final refined ingot. Sampling shall be done on the top and bottom of each ingot.

4.2.2.10.2 Billets and other intermediate steps. Sampling shall be done on the top and bottom of each billet. If the activity performing the testing can identify the billets that came from the top and bottom of the remelted ingot, only billets made from the top and bottom of the remelted ingot need to be sampled.

4.2.2.10.3 Coiled bar, rod, wire, sheet, and strip. Both ends of every coil shall be sampled. If the activity performing the testing can identify the coils that came from the top and bottom of the remelted ingot, only the coils made from the top and bottom of the remelted ingot need to be sampled.

4.2.2.10.4 Straight bar, rod, sheet, strip, and plate. A representative sample of bars, rods, sheets, and plates, taken in accordance with table XXIV, shall be sampled at one end.

4.2.2.10.5 Forgings. Forgings shall be sampled from the intermediate form used to make the forging in accordance with 4.2.2.10.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Sample size for product uniformity testing of straight bar, rod, sheet, and plate.

Lot size	Sample size
1 to 8	All
9 to 15	8
16 to 30	10
31 to 50	12
51 to 85	14
86 to 150	16
Over 150	18

4.2.3 Examination methods.

4.2.3.1 Straightness of rod. In determining straightness in the standard 42-inch distance between supports or in determining straightness when supported at the ends, the rod shall be placed on a precision table equipped with ball bearing rollers and a micrometer or dial indicator. The rod shall then be rotated slowly against the indicator, and the deviation from straightness in any portion of the rod between the supports shall not exceed the permissible variations specified in tables X and XI. The deviation from straightness (throw in one revolution) is defined as the difference between the maximum and minimum readings of the dial indicator in one complete revolution of the rod. Straightness in terms of depth of chord shall be determined with a straightedge, the length of which is not less than that of the piece being measured.

4.2.3.2 Straightness of bar, sheet, and strip. Straightness in terms of depth of chord shall be determined with a straightedge and shall not exceed the permissible variations in 3.9.1.5 and 3.9.3.5, as applicable.

4.2.3.3 Visual and dimensional acceptance criteria. Each sample taken in accordance with 4.2.2.6 shall be subjected to surface inspection for workmanship and dimensions. Any sample lot unit containing one or more visual or dimensional defects shall be rejected.

4.2.3.3.1 Inspection of finish A, hot finished material. Rod or bar surface shall be examined to locate imperfections of maximum severity. These imperfections shall be explored by grinding, filing, or otherwise reducing the cross-section dimension at the site of the imperfection. Sufficient material shall be removed to reach the bottom of the imperfection. The cross-section dimension is measured at this point. If the depth of the imperfection does not exceed the machining allowance in table XXII for the applicable size and form, the product will be acceptable.

4.3 Test procedures.

4.3.1 Chemical composition and product uniformity test. The samples selected in accordance with 4.2.2.1 and 4.2.2.10, as applicable, shall be analyzed by the wet chemical, spectrochemical methods, or combustion techniques to determine conformance with table V. In case of disagreement, the chemical composition shall be determined in accordance with ASTM E1473.

4.3.1.1 Check analysis limits. When the chemistry is determined at a laboratory, other than the laboratory that determined the first chemical analysis, the check analysis limits of ASTM B880 shall apply.

4.3.2 Mechanical tests.

4.3.2.1 Aged material. Mechanical tests required for material shipped in the aged condition shall be done on a sample taken from the material after the final heat treatment. Mechanical tests used to demonstrate the capability of a lot to meet the mechanical test requirements (see 4.3.2.2) may not be used to certify a lot of material shipped in the aged condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.2.2 Unaged material. When mechanical tests are required for material ordered in the unaged condition to demonstrate the capability of the lot to meet the mechanical testing requirements (see 6.2), the test specimens, taken from each lot per 4.2.2.2, 4.2.2.3, 4.2.2.4, and 4.2.2.5, shall be obtained from the material as it is to be to be shipped and heat treated as specified (see 6.2). When no heat treatment is specified, the specimen shall be annealed, if the material to be shipped has not been solution annealed, and aged in accordance with 4.3.2.2.2. Mechanical tests on a lot from the same heat that has previously been tested in the same condition (see 1.2), heat treated as specified above, and found to meet the minimum requirements of this specification may be used to demonstrate the capability of the lot of unaged material to meet the mechanical test requirements.

4.3.2.2.1 Simulated process anneal. When specified (see 6.2 and 6.8.3), the test specimens shall be held at the designated temperature for the designated time to simulate a process anneal. If no time and temperature are specified, the specimens shall be held at 1600 °F for 90 minutes. The simulated process anneal shall be achieved on the test specimens in the as shipped condition (before any other heat treatments). If the material is shipped in the annealed condition, the test specimens shall be re-annealed before they are aged.

4.3.2.2.2 Heat treatment of unaged mechanical test specimens. Unless otherwise specified (see 6.2), age hardening of the test specimens in 4.3.2.2 shall be performed by heating to 1100 to 1125 °F, holding at temperature for 8 to 16 hours depending on hardness of material and desired hardness after aging, furnace cooling to an aim temperature of 900 °F, then air cooling to room temperature.

4.3.2.3 Bar, rod, rectangles, and forged parts. All bar, rod, rectangles, and forged parts shall be tested full-size when practicable. When a machined specimen becomes necessary, enough metal may be removed from the gauge section to meet the limitations of the testing machine or the specimen may be machined to the form and dimensions of the largest standard round specimen in ASTM E8/E8M obtainable from the product being tested. When the product being tested is of such size that an ASTM E8/E8M round specimen less than 0.250 inch diameter is needed, minimum elongation requirements of table I shall not apply.

4.3.2.3.1 Axis of the test specimens. For bar, rod, rectangles, and forged parts up to and including 1½ inches in diameter or thickness, the axis of the test specimen shall coincide with the central axis of the piece; over 1½ inches, the axis shall be located midway between the center and surface of the piece.

4.3.2.3.2 Direction of tensile test specimens. The axis of tension test specimens for bar, rod, rectangles, and forged parts shall be parallel to the direction of rolling, drawing, or flow of metal.

4.3.2.4 Tension test specimens of plate, sheet, and strip. Tension test specimens for plate, sheet, and strip under 0.500 inch thick shall be machined to the form and dimensions of a standard 2-inch gauge length flat in accordance with ASTM E8/E8M. Tension test specimens for plates 0.500 inch and over in thickness shall be machined to the form and dimensions of the largest obtainable standard round in accordance with ASTM E8/E8M.

4.3.2.4.1 Direction of tensile test specimens. Tensile specimens from plate, sheet, and strip shall be taken transverse to the direction of rolling when width will permit. For narrow widths, where a test specimen cannot be obtained from the transverse direction, the test specimen may be taken in the longitudinal direction.

4.3.2.5 Bend test specimens of sheet and strip. Bend tests taken from sheet and strip shall be full thickness longitudinal specimens tested in accordance with ASTM E290. The radius of the pin or mandrel and the angle of bend shall be as specified in table IV.

4.3.2.6 Tension test specimens of wire. Tension test specimens shall be of the full cross-section of the wire and not less than 15 inches in length. Specimens shall be free from sharp bends or kinks. The distance between the jaws of the testing machine, with the specimen in place ready for testing, shall be not less than 10 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.2.7 Wrapping test specimens of wire. Wrapping test specimens shall be full-size sections of wire of a suitable length. Specimens ¼ inch diameter or less, in condition CU (cold drawn [as drawn] condition) or condition AU (cold drawn and annealed), and specimens 0.2294 inch diameter or less, in condition SU (cold drawn, spring temper, as drawn), shall be wrapped around a rod of the same diameter as the wire for eight consecutive close wound turns and examined for cracks by liquid penetrant inspection (see 3.8). Specimens with a diameter greater than 0.2294 inch and less than or equal to ¼ inch, in condition SU (cold drawn, spring temper, as drawn), shall be wrapped around a rod of twice the diameter as the wire for eight consecutive close wound turns and examined for cracks by liquid penetrant inspection (see 3.8). The test specimens shall not have any cracks in them.

4.3.2.8 Yield strength. The yield strength shall be determined by the offset method.

4.3.2.9 Hardness. The hardness shall be determined in accordance with ASTM E10 or ASTM E18, as applicable.

4.3.2.10 Rejection and retests. If any specimen fails to meet the requirements of this specification, the entire lot shall be rejected, subject to the retest provisions specified in 4.3.7.1.

4.3.3 Ultrasonic test.

4.3.3.1 General test requirements.

4.3.3.1.1 Testing requirements. Ultrasonic testing shall be performed in accordance with T9074-AS-GIB-010/271 or MIL-STD-2132, as modified by the requirements specified herein. Testing shall be limited to longitudinal wave and shear wave techniques, as specified herein.

4.3.3.1.2 Surface of rounds. Round rod, 1⅜ inches and under, shall be ordered with a ground or turned surface when ultrasonic testing is specified.

4.3.3.1.3 Surface finish. Finish A, hot finished surfaces of material that will be ultrasonically inspected shall have a roughness average (Ra) of 250 microinches or smoother.

4.3.3.1.4 Acoustic compatibility check. Due to the variable attenuation of nickel-copper-aluminum, an attenuation check shall be performed before each inspection where a reference standard is used. A comparison shall be made between the part to be inspected and the specific reference standard to be used. For longitudinal wave inspections, the attenuation check shall be based on a comparison between a back reflection obtained from similar geometrical areas on the part to be inspected and the reference standard. For shear wave inspections, the attenuation check shall be based on a comparison between a corner reflection signal or a pitch-catch signal obtained from the same total metal distance on the part to be inspected and the reference standard. A minimum of three areas shall be evaluated on the component to be inspected. Differences in attenuation, measured in decibels (dB), shall be compensated for by adding or subtracting the appropriate number of dB from the instrument's gain setting after the calibration reflector height or distance amplitude curve (DAC) is determined and marked on the screen. The most attenuative area evaluated shall be used to determine the attenuation correction value. If the attenuation checks indicate less than a 1-dB difference, no correction is required. When an acoustic compatibility check is performed, the reference standard does not need to meet the signal amplitude comparison test in the acoustic similarity requirements of T9074-AS-GIB-010/271 or MIL-STD-2132.

4.3.3.2 Procedure for ultrasonic inspection of large size bars, rods, and forgings. Coarse grained material may result in an unacceptable ultrasonic signal to perform the required inspections. A single, distinct signal which is clearly discernible from noise shall be used for the inspections. If such a signal is not obtainable using the lowest frequency allowed and with the transducers positioned for the shortest sound path, the material shall be considered un-inspectable and rejected. Inspection shall be performed using either longitudinal wave end scan or shear wave pitch-catch methods as specified in 4.3.3.2.1 and 4.3.3.2.2, respectively. When the procedures of 4.3.3.2.1 are applied to long bars/rods, ultrasonic instruments with high-energy pulses and extended range capability may be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.3.2.1 Longitudinal wave end scan. Inspection shall be performed from the end of the form as illustrated in figure 2. A 1-inch diameter, 2.25-megahertz (MHz) transducer shall be used whenever possible. If the 2.25-MHz transducer does not result in a valid calibration, a 1-inch diameter, 1-MHz transducer may be used. A distance calibration shall be made for the length tested. This calibration shall be accomplished using multiple reflections through the width of the test piece (or acoustically similar reference standard), ensuring that the number of multiples observed is at least equivalent to the length of material to be tested. Alternatively, the distance calibration may be accomplished down the length of the product or an acoustically similar reference standard, using multiples, if necessary, to calibrate for the full length of the part to be tested. Once distance calibration has been established, the inspection shall be performed by setting the signal from the opposite end shall be set to a minimum of 80 percent full screen height. Ensure that the location of the signal from the opposite end represents the form's full length. Both ends of the bar, rod, or forging shall be scanned 100 percent with an overlap of at least 25 percent. Only the reflected longitudinal wave signal shall be used for evaluation as opposed to the mode converted signals which trail the primary longitudinal wave signal.

4.3.3.2.1.1 Validation of inspection method. The ultrasonic inspection method shall be validated by demonstrating the ability to reject a maximum 1-inch diameter flat bottomed hole machined in an acoustically similar bar, rod, or forging with the same or greater diameter as the item to be inspected at a test metal distance at least as long as the item to be inspected (see 4.3.3.4.1.1).

4.3.3.2.2 Shear wave pitch-catch scan. Inspection shall be performed with two scans, 90 degrees apart, along the length of the bar, rod, or forging. For hexagonal bar, three scans are required. The scans shall be directed so that the sound beam passes through the center of the form. The pitch-catch technique illustrated in figure 3 shall be used. The maximum dimension of the transducer active element shall not exceed 1 inch. The nominal transducer frequency shall be a minimum of 1 MHz. The pitch (sending) transducer and the catch (receiving) transducer shall be from the same manufacturer and of the same type, size, and frequency. The sound beam angle shall be as shown in figure 3 (±2 degrees). The 41-degree transducers may be used for the end scan procedure, still using the specified ¼-inch grid lines if their use results in an acceptable signal presentation (the 41-degree transducers typically provide enough 45-degree sound energy to perform the required inspections).

4.3.3.2.2.1 Center scan. The catch transducer shall be placed in position R1 or R2 (see figure 3) for the center scan depending on the ease of fixturing and the ability to penetrate highly attenuative material. The transducers for the center scan shall be fixtured and locked in position during scanning. The fixture shall be adjustable to compensate for different sizes and to peak the signal during initial set up. With the transducers aligned and locked into place, the transmitted signal shall be set between 80 and 90 percent full screen height. The transducers shall be scanned along the entire length of the bar, rod, or forging. Changes in material attenuation will be shown by gradual change in signal amplitude and will occur gradually along the length of the form, either highest or lowest in the center and gradually increasing or decreasing as the fixture is scanned toward the bar ends. Gradual changes in signal amplitude shall be compensated for with the instrument gain control to maintain 50 to 90 percent signal along the entire length of the bar, rod, or forging.

4.3.3.2.2.2 End scan. The end of the bar, rod, or forging shall be scanned with a similar technique. For end scans, fixturing is not practical and an indexing system shall be used to track transducer positions. As illustrated in figure 4, the end of the form shall be marked at ¼-inch increments across the cross-section. One side of the form shall be similarly marked in ¼-inch increments for a length equal to the cross-section or thickness. The transducers shall be initially placed nose to nose at the corner of the form and an 80 to 90 percent full screen height signal shall be obtained. The transducers shall then be indexed one mark at a time simultaneously and the signal read for each index position. It is mandatory to hold the transducers at their index positions and then read the signal. Do not peak the signal for the maximum signal amplitude. Changes in attenuation due to the changing sound path and changes in material attenuation will be shown by a gradual change in signal amplitude while the transducers are indexed. Gradual changes in signal amplitude shall be compensated for with the instrument gain control to maintain a 50 to 90 percent signal along the entire scan length.

4.3.3.3 Procedure for additional, elective ultrasonic inspections. When specified (see 3.7.1.2), rod, bar, ring, hollow round products, disc and pancake forgings, and plate shall be ultrasonically inspected in accordance with T9074-AS-GIB-010/271 or MIL-STD-2132. Sheet, strip, and wire are specifically excluded from these requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.3.3.1 Ring and hollow round (pipe and tube) products. In addition to the circumferential scan required by T9074-AS-GIB-010/271 or MIL-STD-2132, products shall be inspected with a longitudinal wave test directing the sound beam radially and axially.

4.3.3.3.2 Plate. Calibration shall be performed on a flat bottomed hole in lieu of the plate back reflection specified in T9074-AS-GIB-010/271 or MIL-STD-2132. For plate thicknesses up to and including 4 inches, a ¼-inch diameter hole shall be used. For plate thicknesses greater than 4 inches, a ½-inch diameter hole shall be used. The hole shall be drilled into the test piece or into a separate defect-free specimen of the same size (within ±⅛ inch), shape, and material (or acoustically similar material). The hole shall be drilled to a depth of one-half the thickness or 1 inch, whichever is less. Plate shall be scanned continuously on either a 24-inch grid with one diagonal on each grid, or a 12-inch grid.

4.3.3.3.2.1 Reference notch removal. If reference notches or flat bottomed holes are made in the material to be tested, they shall be so located that their subsequent removal will not impair the suitability of the material for its intended use.

4.3.3.4 Acceptance criteria.

4.3.3.4.1 For ultrasonic inspection of large size bars, rods, and forgings (forms 2 and 7).

4.3.3.4.1.1 Longitudinal end scan. A 90 percent or greater loss of back reflection signal (except when the transducer is within one transducer diameter of the side of the bar) or any discontinuity signal greater than 25 percent of the established reference level shall be cause for rejection.

4.3.3.4.1.2 Shear wave pitch-catch scan. Abrupt reductions in signal amplitude (6 dB or more in 1 inch or less of length) shall be cause for rejection.

4.3.3.4.2 Additional, elective ultrasonic inspections.

4.3.3.4.2.1 Shear wave. Any material that produces indications equal to or larger than the response from the reference notch or higher than the straight line joining the two peak amplitudes shall be rejected.

4.3.3.4.2.2 Longitudinal wave. Any material that produces indications equal to or larger than the response from the reference hole, or contains areas which result in a 90 percent or greater loss of back reflection (except when the transducer is within one transducer diameter of the side of the bar), shall be rejected. Material shall be tested using a square, rectangular, or circular transducer having an effective area of 1 square inch or less, but no dimensions shall be smaller than the diameter of the reference hole. In the event of disagreement on the degree of back reflection loss, it shall be determined by the contact method using a 1- to 1⅛-inch diameter transducer or one whose area falls within this range.

4.3.4 Liquid penetrant inspection.

4.3.4.1 Procedure. Liquid penetrant inspection shall be in accordance with T9074-AS-GIB-010/271 or MIL-STD-2132.

4.3.4.2 Surface requirements. The surface produced by hot working is not suitable for liquid penetrant testing. Therefore, liquid penetrant testing will not be applicable to products ordered with a hot finished surface.

4.3.4.3 Acceptance criteria. Linear defects revealed by liquid penetrant inspection shall be explored by grinding or other suitable means. Depth of defects shall not exceed the dimensional tolerance of the material.

4.3.5 Grain size determination. When required (see 3.5), the sample selected for grain size determination shall be macroetched to reveal the grain structure. The sample shall be prepared so that the surface to be etched has a suitable finish for macroetching and be etched in accordance with ASTM E340. The macroetched surface shall be visually examined (unaided visual or low power, up to 10 times magnification). One grain size specimen shall be taken from the coarsest grained area. The selected grain size specimen shall be prepared and inspected in accordance with ASTM E112. The inspected surface of the specimen shall be perpendicular to the major direction of working.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.6 Slow strain rate tensile test procedure. The slow strain rate tensile test will determine if a lot of metal is susceptible to intergranular cracking. The slow strain rate test specimens shall be heat treated using the final heat treatment procedure used on the lot before being placed into service. Any lot that has been rejected shall not be submitted for acceptance.

4.3.6.1 Heat treatment of slow strain rate test specimens. Specimens representing lots of material sold in the annealed condition shall be heat treated as specified (see 6.2). Lots of material that are aged or re-solution annealed and aged with a heat treating procedure that is not equivalent (see 4.3.6.1.1) to the procedure used on the slow strain rate tensile test shall be re-tested using specimens taken from the material after the final heat treatment.

4.3.6.1.1 Slow strain rate tensile test specimen heat treatment. For the purpose of sampling slow strain rate tensile tests in accordance with 4.2.1.3, two heat treatments shall be considered to be equivalent if the following conditions hold:

a. If the material is solution annealed, the solution annealing temperatures are between 1700 and 1900 °F or, if above 1900 °F, are within 50 °F of each other.

b. The age hardening temperatures for the highest temperature age hardening step are within 50 °F of each other. Variations in the aging time and the cooling method under 1000 °F are allowed.

c. Variations in aging times, variations in aging temperatures other than the highest-temperature aging step, and variations in cooling rates between aging steps, after the last aging step, do not affect the definition of equivalence for purposes of determining slow strain rate test sampling frequency.

4.3.6.2 Slow strain rate tensile test specimen dimensions. Each specimen selected in accordance with 4.2.2.2 shall be machined to make either a small or a large notched test specimen. The notched test specimens shall be in accordance with figure 1 and the following:

a. Any loading fixture may be used.

b. Cylindrical surfaces and specimen heads shall be machined with an eccentricity with respect to the notch not exceeding 0.001 inch.

c. The notch root radius (R) shall be 0.002±0.0005 inch. The notch shall have an opening of 60±5 degrees.

4.3.6.2.1 Small notched test specimen. The diameter (D) of the small notched test specimen shall be 0.500±0.005 inch and the diameter at the notch (d) shall be 0.353±0.005 inch. The test section (L) of the small notched test specimen shall be at least 1 inch long.

4.3.6.2.2 Large notched test specimen. The diameter (D) of the large notched test specimen shall be 1.060±0.005 inches and the diameter at the notch (d) shall be 0.750±0.005 inch. The test section (L) of the large notched test specimen shall be at least 2.13 inches long.

4.3.6.3 Slow strain rate tensile test. Each slow strain rate tensile specimen (see 4.3.6.2) shall be tested in accordance with ASTM E8/E8M at a temperature of 700±2.5 °F. The displacement rate shall not exceed 0.001 inch per inch of the distance between the grips per minute. Alternatively, if an extensometer is used to measure displacement, the displacement rate for the specimens shall not exceed 0.001 inch per inch per minute. Loading below 50 percent of the minimum yield strength of the material can be conducted at any rate under 100,000 pounds per square inch (psi) per minute.

4.3.6.4 Grain size determination. The average grain size of the half of the slow strain rate tensile specimen not used for fracture surface analysis in accordance with 4.3.6.5 shall be determined as specified in ASTM E112. The grain size shall be measured such that the average grain size in micrometers is determined. The grain size shall be taken parallel to the fracture surface and as close to the fracture surface as practical.

4.3.6.5 Fracture surface analysis. After fracture, the fracture surface of each specimen shall be examined and the fracture mode characterized as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.3.6.5.1 Fracture surface photography. A series of eight photographs shall be taken of the fracture surface at equally spaced intervals of 45 degrees around the specimen circumference, with the first site selected randomly. Each photograph shall be taken using a scanning electron microscope (SEM) at a magnification scaled to the grain size using the relationship below:

$$M = 10{,}000/G$$

Where M is the magnification and G is the grain size in micrometers (μm). If the calculated magnification is not a preset level on the SEM being used, the magnification shall be decreased to the nearest preset level. For grain sizes larger than 15 micrometers, a small portion of the machined notch root must be visible in each photograph. For grain sizes smaller than 15 micrometers, the center of each photograph must be approximately 150 micrometers from the notch root.

4.3.6.5.2 Determination of intergranular fracture sites. The photographs of 4.3.6.5.1 shall be used to determine intergranular fracture features. An intergranular fracture feature shall be defined as two adjacent and contiguous facets demarcated by a distinct grain boundary or three adjacent and contiguous facets demarcated by grain boundaries that come together at a triple point. The facets shall be clearly identifiable as a grain surface; otherwise, they shall not be considered in defining a site of intergranular fracture.

4.3.6.5.3 Quantification of intergranular fracture sites. The template illustrated in figure 5 shall be transferred to a transparency that shall be placed over each of the nominal 3½- by 4½-inch photographs obtained in accordance with 4.3.6.5.1. Intercepts with intergranular fracture features with the circles on the template shall be counted for each photograph. An intercept shall be defined as the intersection of one intergranular fracture feature with a circle on the template. If one intergranular fracture feature extends over several circles on the template, it shall be counted as one feature. The intercepts from all of the photographs of each fracture surface shall be totaled.

4.3.6.5.4 Testing laboratory. When specified (see 6.2), the slow strain rate tensile test shall be performed at the laboratory specified by the purchaser.

4.3.6.6 Acceptance criteria. The total number of intercepts counted in 4.3.6.5.3 for any fracture surface evaluated shall not exceed 100.

4.3.6.7 Rejection and retests. If any specimen fails to meet the requirements of this specification, the entire lot shall be rejected, subject to the retest provisions specified in 4.3.7.1.1.

4.3.7 Replacement, retest, rejection, and resubmittal.

4.3.7.1 Mechanical property and slow strain rate retests.

4.3.7.1.1 Discarded tests. If any test specimen shows defective machining or reveals flaws which cause inaccurate test results, it may be discarded and a replacement test specimen substituted. If any test specimen has not been tested in accordance with the requirements of this specification, it shall be discarded and a suitable replacement test specimen substituted.

4.3.7.2 Rejection. Unless otherwise specified (see 6.2 and 6.4 to 6.4.1.), when any test specimen, except those test specimens discarded in accordance with 4.3.7.1.1, is not in accordance with specification requirements for the characteristics being tested, the lot represented by the specimen shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with ASTM D3951, unless detailed requirements are specified in the contract (see 6.2).

6. NOTES

(This section contains information of a general or explanatory in nature that is helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 General. Nickel-copper-aluminum alloy, UNS N05500 is used where a combination of high strength and resistance to corrosion are needed. For optimum tool life, the annealed condition is recommended. Cold drawn material is intended for automatic machining operations. Material solution annealed and aged to a hardness range of 24-35 HRC is intended for seawater applications where toughness is required.

6.1.2 Annealed sheet and strip. Nickel-copper-aluminum alloy sheet and strip in the annealed condition may be used in the manufacture of articles in which other operations, including but not limited to, heading, bending, curling, drawing, forming, lock-seaming, and welding are used.

6.1.3 Cold rolled strip. Cold rolled strip with a half hard or a full hard temper are used, where light deformations or liberal bending radii are involved, for applications such as springs.

6.1.4 Annealed wire. Nickel-copper-aluminum alloy wire in the annealed condition is intended for severe forming, cold heading, and roll threading operations.

6.1.5 Spring temper wire. Nickel-copper-aluminum alloy wire, spring temper in the as drawn condition, is intended for coiling of helical springs which are to be aged hardened after coiling in order to obtain maximum strength and hardness. Where helical springs are to be cold pressed, the cold pressing is to be done after coiling and aging.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Form, condition, and surface finish of material required (see 1.2 and 1.2.2).
c. When a forging sketch is required (see 3.1.1).
d. When MIL-STD-1684 may be used instead of AMS 2750 (see 3.1.2).
e. Size required (see applicable table)..j
f. When a product uniformity test is required (see 3.2).
g. When hardness measurements may be utilized for forged rings and discs (see 3.3.3).
h. When unaged bar and rod (form 1) are required to have a fine grain size (see 3.5.1).
i. If additional, elective ultrasonic inspection is required (see 3.7.1.2).
j. If optional liquid penetrant inspection is required (see 3.7.2).
k. When dimensional tolerances other than those specified are required (see 3.9; table VII, note 2/; table XIV, note 6/; table XVI, note 5/; and table XVII, note 1/).
l. When dimensional tolerances are other than specified (see 3.9).
m. Lengths required and whether cut-to-length, random mill lengths, nominal (stock) lengths, or multiples of specified lengths (see 3.9.1.3.1, 3.9.3.3.1, and 3.9.3.3.2).
n. If multiples of a specified length are ordered, whether a ¼-inch allowance should be made for each cut (see 3.9.1.3.1).
o. If edges are other than specified (see 3.9.1.4, 3.9.3.4.1, 3.9.3.4.2, and 3.9.3.4.3).
p. Forging dimensions (see 3.9.2).
q. If marking should be other than specified (see 3.10).
r. When mechanical tests are required for rod and bar intended for reforging (see 4.3.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

s. If mechanical testing is to be done on material ordered in the unaged condition to demonstrate its capability of being aged to meet the required mechanical properties and, if desired, the heat treating procedure to be used on the test specimens (see 4.3.2.2 and 4.3.2.2.2).

t. If a simulated process anneal is required on heats given capability testing (see 4.3.2.2.1), and specific time and temperature for the simulated process anneal.

u. The final heat treatment procedure to be used on the lot before being placed into service and on the slow strain rate tensile test specimens (see 4.3.6.1).

v. When the laboratory conducting the slow strain rate tensile test is selected by the purchaser (see 4.3.6.5.4).

w. When rework is permitted (see 4.3.7.2).

x. Packaging, packing, and marking requirements (see 5.1).

6.3 Certification of quality conformance. Certification of quality conformance should provide quantitative results and requirements for all specified chemical, grain size, slow strain rate tensile, and mechanical tests. A slow strain rate tensile test report should contain the temperatures, holding times, and cooling methods of the last heat treatment procedures used on the material when the slow strain rate tensile test was taken from it and all the heat treatment procedures used on the test specimens; the average grain size of the test specimens; the displacement rate; the notch tensile strength; and the photographs, with transparencies, marked to show the intercepts counted as intergranular fracture features, the photomagnification of the photographs, and the number of intercepts counted. For material supplied in the unaged condition, the certification should prominently display the words: "AS-SUPPLIED MATERIAL CONDITION IS UNAGED", and include the statement: "This certifies that the material contained in this lot, when properly age hardened, is capable of meeting the mechanical properties for the corresponding age hardened condition specified in QQ-N-286."

6.4 Reinspection of rejected lots.

6.4.1 Rejected lots.

6.4.1.1 Mechanical tests. A lot rejected for failing mechanical tests may be resubmitted for acceptance testing only after rework is performed to correct the nonconforming condition without adversely affecting the conforming properties. All mechanical and slow strain rate testing, including those which were initially conforming, should be repeated after rework.

6.4.1.2 Dimensional tests. A lot rejected for failing visual or dimensional tests that has been reworked to bring it into conformance with the requirements of the specification may be resubmitted for acceptance testing provided it does not adversely affect the conforming properties. Mechanical and slow strain rate tests should be repeated only when the lot is re-heat treated.

6.4.1.3 Individual pieces. When a lot, rejected for failing visual or dimensional inspection, consists of more than one piece, each remaining piece in the lot may be resubmitted for testing for the nonconforming characteristic and each piece that conforms to all specification requirements may be considered for acceptance.

6.5 Alternate ultrasonic inspection procedure. Consideration may be given to alternate ultrasonic inspection procedures when reviewed and approved by the purchaser.

6.6 General information.

6.6.1 Finishes of age hardened material. Age hardened material is typically supplied with a thin, dark oxide finish when the material is in Finish A, hot finished (see 1.2.2.a); Finish B, hot finished forging stock (see 1.2.2.b); and Finish C, as drawn (see 1.2.2.c). The oxide layer is normally removed from Finish D, rough turned or rough ground rounds (see 1.2.2.d); Finish E, semi-smooth machined rounds (see 1.2.2.e); Finish F, smooth finished machined rounds (see 1.2.2.f); Finish G, bright finish (see 1.2.2.g); and Finish H, pickled (see 1.2.2.h).

6.6.2 Allowances on hot finished material. When ordering hot finished material, allowances should be made as listed in table XXII to allow for clean-up to the user's finished cross-sectional dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

6.7 Hot working cautions.

6.7.1 Cooling. Material should not be cooled slowly following hot working operations. Slow cooling will result in self-hardening as cooling takes place through the precipitation hardening range. Generally, sections ¾ inch thick and smaller will air cool or can be cooled with an air blast rapidly enough to prevent self-hardening, provided the parts are not piled one on top or against each other. Sections over ¾ inch are normally water quenched.

6.7.2 Interruption of forging operation. Should a forging operation on this material be interrupted to the extent that the heated bars are likely to cool below 1450 °F, the material should be quenched rather than be permitted to air cool below that temperature.

6.7.3 Heating recommendations. With reference to the heating times shown herein, care should be taken to obtain the proper temperature throughout the cross-section on large diameters or thicknesses. Also, it should be understood that material should be brought to annealing temperature or hot working temperature as rapidly as possible to avoid aging of the material on its rise in temperature; otherwise, cracking or thermal splitting of the material might occur, particularly in subsequent hot working operations.

6.7.4 Maximum forging temperature. This material is usually hot worked at a maximum temperature of 2100 °F. Heating to higher temperatures could promote excessive grain growth and oxidation.

6.8 Heat treatment guidance. There is no one prescribed heat treatment for this alloy. Times and temperatures are approximate and may be modified to account for the size, shape, microstructure, and condition of the metal to be heat treated.

6.8.1 Annealing.

6.8.1.1 Process annealing. If, for any reason, it is desired to soften hard material during fabrication, this may be accomplished by holding at 1600 to 1900 °F as necessary to provide for recrystallization. The time at temperature should be commensurate with section thickness and followed by water quenching. The total time at temperature should be limited to avoid the formation of titanium carbides. The material should be held at temperature for the minimum amount of time needed to dissolve any phases that may have formed in the material during previous processing. It is recommended that the total cumulative time of exposure to in-process softening and solution annealing temperatures for all of the annealing cycles should be less than 1½ hours. Nickel-copper-aluminum alloy with excessive quantities of titanium carbides will have a weaker response to the aging heat treatment and may not be capable of meeting the minimum mechanical property requirements when aged. Titanium carbides are stable at normal solution annealing and will not be re-dissolved into the material.

6.8.1.2 Final solution anneal. Unaged material is usually final solution annealed by the manufacturer and should not need to be solution annealed again. When final solution annealing a cold worked product, a temperature of 1900 °F should be used followed by water quenching. When final solution annealing hot worked product and product in any other condition, a temperature of 1800 °F should be used followed by water quenching. The material should be held at temperature for the minimum amount of time needed to dissolve any phases that may have formed in the material during previous processing. Heating and cooling times should be kept to a minimum to avoid precipitation of detrimental phases.

6.8.2 Age hardening. Aging (hardening) may be accomplished by holding the metal at 1100 to 1125 °F for 8 to 16 hours followed by furnace cooling to 900 °F at a rate of 15 to 25 °F per hour and then air cooling. Age hardening may also be accomplished by holding the metal at 1100 °F for a period of 16 hours, furnace cooling to 1000 °F, holding for a period of 6 hours, furnace cooling to 900 °F, holding for a period of 8 hours, and air cooling.

6.8.3 Simulated process annealing. The simulated process anneal (see 4.3.2.2.1) is intended to be used for fastener and other product manufacturers that intend to subject the metal to one or more process anneals. Simulated process annealing the test specimens will simulate the detrimental effects of process annealing and allow the manufacturer to ensure that their final product will meet the minimum mechanical properties when the material is aged. Nickel-copper-aluminum alloy that has been exposed to excessive process annealing temperatures may not be capable of passing the mechanical test requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

6.8.4 Recovering or refining process annealed material. Titanium carbides that are formed during process annealing can be dissolved by solution annealing the product for 30 minutes at a minimum temperature of 2050 °F. Nickel-copper-aluminum alloy that has been heated at temperatures over 2000 °F will have a large grain size and reduced formability. The large grain size may be reduced by cold working the material, re-solution annealing the material at 1800 °F, and then age hardening the material. If properly done, this will cause it to recrystallize to a finer grain size and enhance the material's response to the age hardening treatment. This technique may also be used to produce a product with a fine uniform grain size.

6.9 Definitions.

6.9.1 Procuring activity. The procuring activity is any activity that procures the product from the primary melter or another procuring activity to further process or use the product.

6.9.2 Product check analysis. A product check analysis is any chemical analysis done by any procuring activity. The product check analysis is intended to allow the procuring activity to confirm that the product meets the mandatory chemical requirements of this specification. Allowances to the chemistry requirements in table V are given in ASTM B880 to accommodate the differences in results expected when different laboratories test the same product. The only difference between product check analysis and product uniformity testing is the number and location of the samples taken.

6.9.3 Product uniformity test. A product uniformity test is intended to confirm that the product is uniform in composition and that the entire product shipped is in compliance with the chemical requirements of this specification. Improper processing of the product can result in a portion of the product having a chemistry that will not allow the product to achieve the required minimum mechanical properties when it is properly solution annealed and aged. The only difference between product check analysis and product uniformity testing is the number and location of the samples taken.

6.10 Subject term (key word) listing.

Bars

Forgings

Rods

Ultrasonic

6.11 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

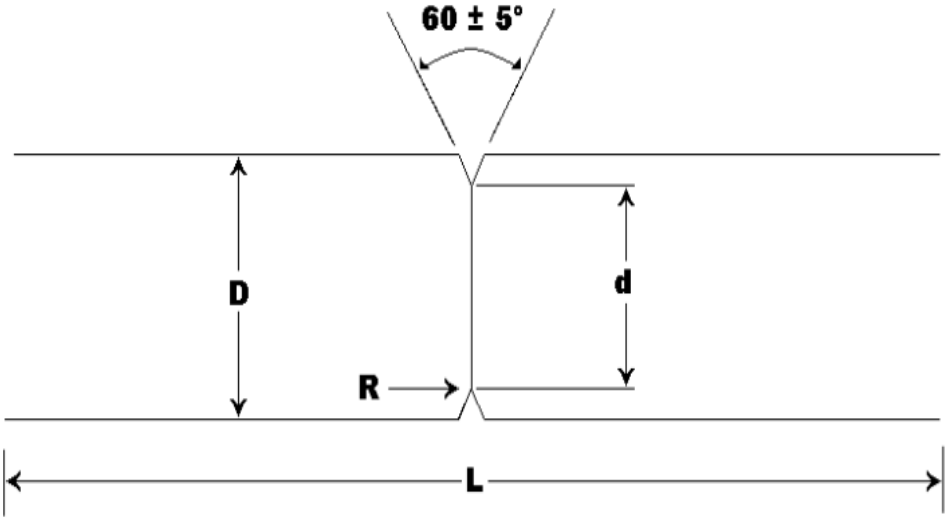
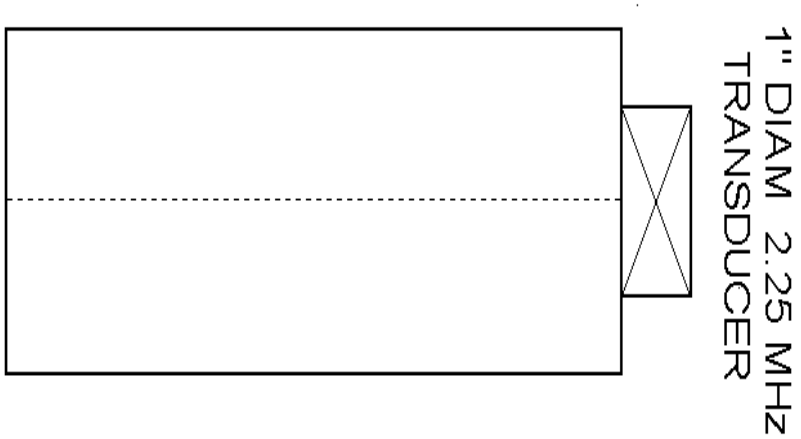


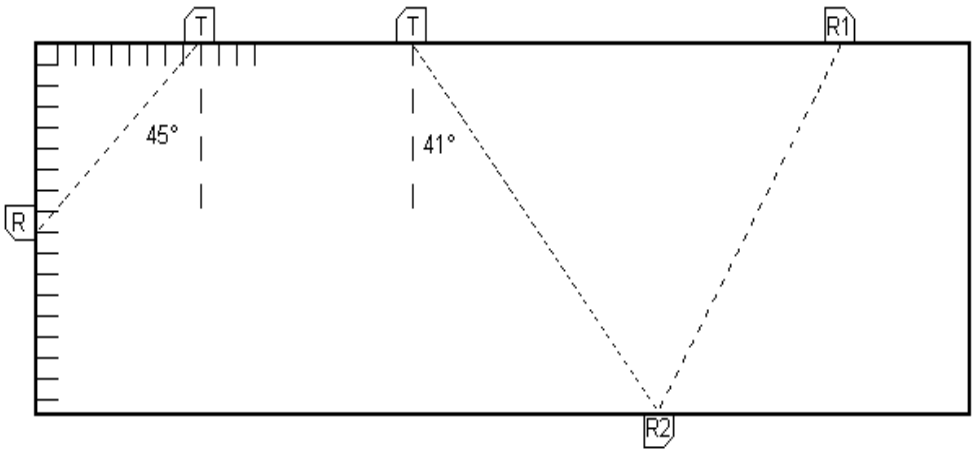
FIGURE 1. Dimensions of the notched test specimen.



NOTE: This drawing is for illustrative purposes only and is not to scale

FIGURE 2. Longitudinal end scan inspection technique for large size bar, rod, and forgings.

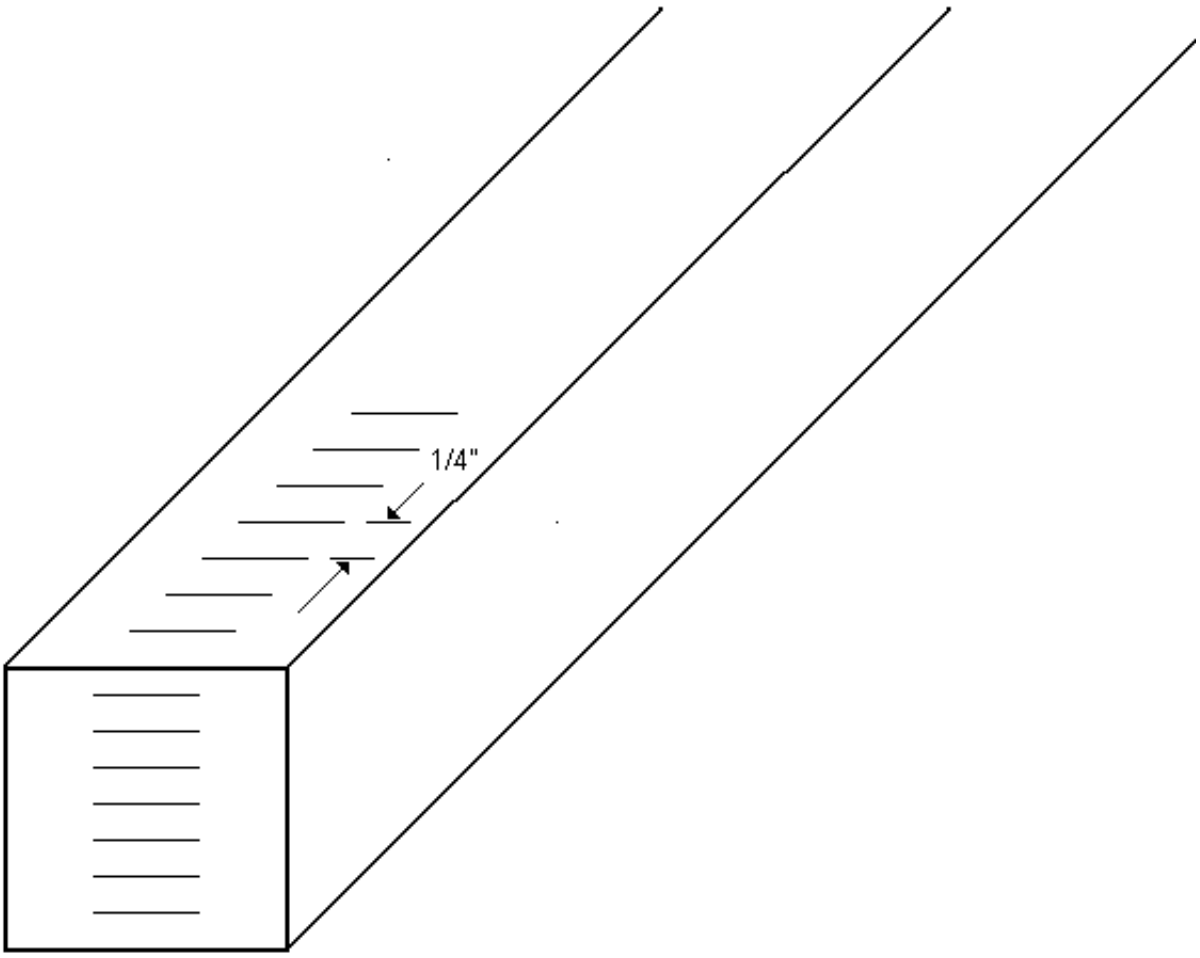
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.



NOTE: This drawing is for illustrative purposes only and is not to scale (see 4.3.3.2.2).

FIGURE 3. Shear wave pitch-catch technique for large bar, rod, and forgings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.



NOTE: This drawing is for illustrative purposes only and is not to scale (see 4.3.3.2.2.2).

FIGURE 4. Index marking layout for shear wave pitch-catch end scan of large bar, rod, and forgings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

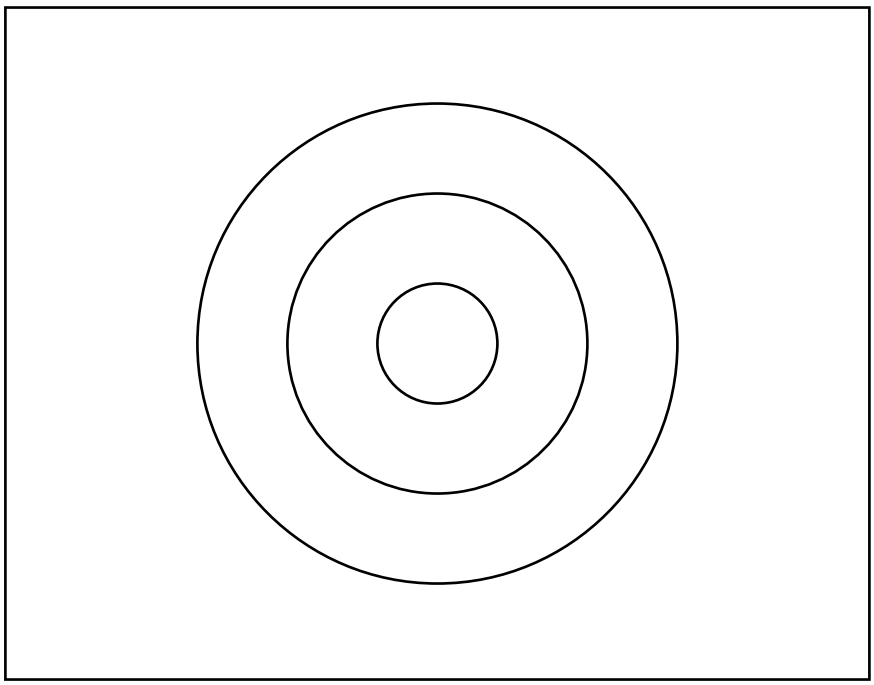


FIGURE 5. Template for slow strain rate fracture test evaluation (concentric circles of 2.5000-, 1.5625-, and 0.625-inch diameter in a 3.5- by 4.5-inch frame).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS

Custodians:
Army – MR
Navy – SH
Air Force – 20

Review Activities:
Army – AV, CR, GL
Navy – OS
Air Force – 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy − SH
(Project 9525-2013-001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.